DAY 12 (3/2/26)-HACKERRANK PROBLEMS

1. Write a program to accept a array elements from the user and display the smallest element in the array

Input Format

Accept an array elements from the user

Constraints

nil

Output Format

Display the smallest element in the array

Sample Input 0

5

10 20 30 40 50

Sample Output 0

10

Program:

```
import java.io.*;

import java.util.*;

import java.text.*;

import java.math.*;

import java.util.regex.*;


public class Solution {


    public static void main(String[] args) {


        Scanner sc=new Scanner(System.in);

        int n=sc.nextInt();

        int a[]=new int[n],i;
```

```
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        int min=a[0];
         for (i = 0; i < n; i++) {
            if (a[i] < min) {
                min = a[i];
            }
        }

        System.out.print(min);
    }
}
```

2. Write a program to accept the array and position in which the new element to be inserted from the user and display the array.

Input Format

number of elements

array elements

element to be inserted

position where insertion should be done

Constraints

NIL

Output Format

Updated array

Sample Input 0

10

1 2 3 4 5 6 7 8 9 10

5

2

PROGRAM:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int i;
        int a[]=new int[n+1];
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        int e=sc.nextInt();
        int pos=sc.nextInt();
        for(i=n;i>=pos;i--){
            a[i]=a[i-1];
        }
        a[pos-1]=e;
        for(i=0;i<n+1;i++){
        System.out.print(a[i]+" ");
        }
    }
```

}

3. Write a program to accept the array elements ,position from the user and delete the element at the given position

Input Format

number of elements

array elements

position to be deleted

Constraints

nil

Output Format

updated array

Sample Input 0

5

1 2 3 4 5

6

Sample Output 0

Deletion not possible.

Sample Input 1

10

1 2 3 4 5 6 7 8 9 10

1

Sample Output 1

2 3 4 5 6 7 8 9 10

PROGRAM:

```
import java.io.*;

import java.util.*;

import java.text.*;

import java.math.*;

import java.util.regex.*;
```

```
public class Solution {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int i;
        int a[]=new int[n];
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }

        int pos=sc.nextInt();

        pos = pos - 1;

        if (pos < 0 || pos >= n) {
            System.out.print("Deletion not possible.");
        } else {
            for (i = pos; i < n - 1; i++) {
                a[i] = a[i + 1];
            }
            for (i = 0; i < n - 1; i++) {
                System.out.print(a[i] + " ");
            }
        }

    }
}
```

4. Write a program to accept the array elements from the user and find the largest element in the array.

Input Format

Number of elements

Array elements

Constraints

NIL

Output Format

Largest element in the array

Sample Input 0

10

10 20 30 40 50 60 70 80 90 100

Sample Output 0

100

PROGRAM:

```
import java.io.*;

import java.util.*;

import java.text.*;

import java.math.*;

import java.util.regex.*;


public class Solution {


   public static void main(String[] args) {

      /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class should be named Solution. */

      Scanner sc=new Scanner(System.in);

      int n=sc.nextInt();

      int a[]=new int[n],i;

      for(i=0;i<n;i++){
```

```
            a[i]=sc.nextInt();


        }
        int max=a[0];
        for(i=0;i<n;i++){
            if(max<a[i]){
                max=a[i];
            }


        }
        System.out.print(max);


    }
}
```

5. Write a program to accept the array from the user and display all the unique elements in the array

Input Format

Accept a array size and array elements

Constraints

nil

Output Format

Display the unique element

Sample Input 0

10

1 2 2 3 3 4 4 5 5 10

Sample Output 0

1 10

PROGRAM:

```
import java.io.*;
import java.util.*;
```

```
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int i,j;
        int a[]=new int[n];
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
         boolean u =false;
           for ( i=0;i<n; i++) {
            int count=0;
            for ( j=0;j<n;j++) {
                if (a[i]==a[j]) {
                    count++;
                }
            }
            if (count==1) {
                System.out.print(a[i] + " ");
                u=true;
            }
        }
        if(!u) {
```

```
            System.out.println("No unique elements in the array");


    }
}}
```

6. Given an array of elements, write a program to find the K th smallest element in the array

Input Format

number of elements

k value

array elements

Constraints

NIL

Output Format

k th smallest element

if k value is greater than n print "Out of Range"

Sample Input 0

8

6

32 4 65 78 3 21 68 1

Sample Output 0

65

Sample Input 1

5

10

1 2 3 4 5

Sample Output 1

Out of Range

PROGRAM:

```
import java.io.*;
```

```
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
       int n = sc.nextInt();
       int k = sc.nextInt();
       int[] arr = new int[n];
       for (int i = 0; i < n; i++) {
           arr[i] = sc.nextInt();
       }
       if (k> n || k <= 0) {
           System.out.println("Out of Range");
           return;
       }
       Arrays.sort(arr);
       System.out.println(arr[k - 1]);
    }
}
```

7. Write a program to Rotate Array n times

Input Format

Accept array size,array elements and key value

Constraints

nil

Output Format

Display the rotated array

Sample Input 0

8

1 2 3 4 5 6 7 8

6

Sample Output 0

3 4 5 6 7 8 1 2

PROGRAM:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int a[]=new int[n],i;
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();

        }
        int k=sc.nextInt();
        k=k%n;
        k=n-k;
        for (i=k;i<n;i++) { //6
            System.out.print(a[i] + " ");
        }
```

```
        for (i=0;i<k;i++) {
            System.out.print(a[i] + " ");
        }


    }
}
```

8. Write a C program to Find common elements from 3 sorted array

Input Format

Accept array elements

Constraints

nil

Output Format

common element from the sorted array

Sample Input 0

6 3 4

1 3 4 5 6 7

1 2 5

1 3 5 7

Sample Output 0

1 5

PROGRAM:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;


public class Solution {
```

```
public static void main(String[] args) {
    Scanner sc=new Scanner(System.in);
    int n1=sc.nextInt();
    int n2=sc.nextInt();
    int n3=sc.nextInt();
    int a[]=new int[n1];
    int b[]=new int[n2];
    int c[]=new int[n3],i,j,k;
    for(i=0;i<n1;i++){
        a[i]=sc.nextInt();
    }
    for(i=0;i<n2;i++){
        b[i]=sc.nextInt();
    }
    for(i=0;i<n3;i++){
        c[i]=sc.nextInt();
    }
    boolean u = false;
    for(i=0;i<n1;i++){ //1
        int count = 0;
        for(j=0;j<n2;j++){//1
        for(k=0;k<n3;k++){//1 //3 //5 //7
        if(a[i]==b[j] && b[j]==c[k]){
            count++;
        }
    }
    }
    if (count == 1) {
    System.out.print(a[i] + " ");
```

```
        u = true;
        }
        }
        if (!u) {
            System.out.println("No unique elements in the array");
        }
    }
}
```

9. Write a program to accept the array and search the given element present in the array

Input Format

Accept the number of array elements,array and elements to be searched as an input

Constraints

nil

Output Format

Found/Not found

Sample Input 0

10

1 2 3 4 5 6 7 8 9 10

5

Sample Output 0

Found

PROGRAM:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {
```

```
    public static void main(String[] args) {
        /* Enter your code here. Read input from STDIN. Print output to STDOUT. Your class
should be named Solution. */
        Scanner sc=new Scanner(System.in);
        int n=sc.nextInt();
        int found=0;
        int a[]=new int[n];
        int i;
        for(i=0;i<n;i++){
            a[i]=sc.nextInt();
        }
        int k=sc.nextInt();
        for(i=0;i<n;i++){
            if(k==a[i]){
                found++;
            }
        }
        if(found==1){
            System.out.print("Found");
        }
        else{
            System.out.print("Not found");
        }
    }
}
```

10. Write a program to accept the array from the user and right rotate second half of the array by n times

Input Format

Accept the number of elements , array and number of times need to be rotated as input

Constraints

NIL

Output Format

Rotate half of the array by n times

Sample Input 0

10

1 2 3 4 5 6 7 8 9 10

4

Sample Output 0

1 2 3 4 5 10 6 7 8 9

Sample Input 1

5

1 2 3 4 5

6

Sample Output 1

1 2 3 4 5

PROGRAM:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;
import java.util.*;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
```

```
        int[] a = new int[n];

        for(int i=0;i<n;i++)
            a[i] = sc.nextInt();

        int r = sc.nextInt();

        int start = (n%2==0) ? n/2 : (n+1)/2;
        int len = n - start;

        if(r < len){
            int last = a[n-1];
            for(int i=n-1;i>start;i--)
                a[i] = a[i-1];
            a[start] = last;
        }

        for(int i=0;i<n;i++)
            System.out.print(a[i] + " ");
    }
}
```